

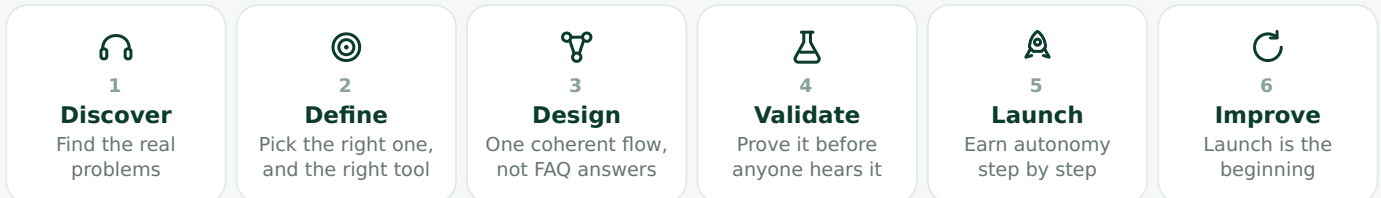
From merchant problem to working AI: how I would tackle it

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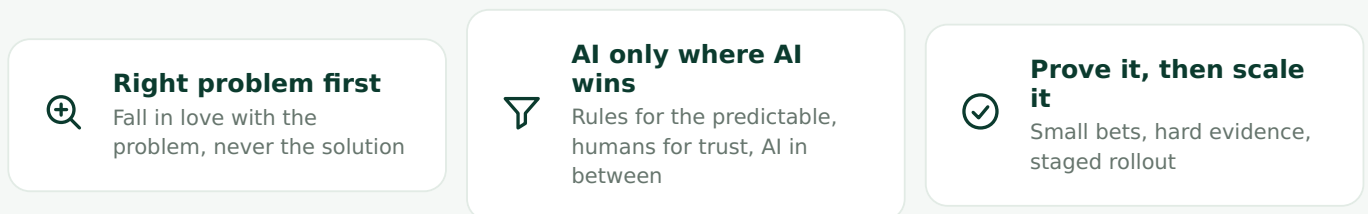
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A method walkthrough with one running example from Flatpay's world: **"My terminal won't take payments."** Six phases, one page each. Built without inside knowledge: every assumption is cheap to correct.

THE METHOD: SIX PHASES



THREE BELIEFS BEHIND IT



THE RUNNING EXAMPLE

"My terminal won't take payments"

A stressed merchant, customers in line, revenue stopped. The most urgent call type in merchant support, and the example that runs through every phase of this document: from hearing it on live calls to measuring whether the AI truly fixed it.



WHAT THE SIX PHASES DELIVER



PHASE 1 · DISCOVER

Understand how merchants actually reach for help

"You can't design the fix before you've heard the problem in the merchant's own words."

FOUR WAYS IN, BEFORE ANY SOLUTION TALK



Listen to live calls

How help is asked for in the moment, not described afterwards



Interview the experts

Support leads and product specialists, market by market



Mine the data

Tickets, transcripts and telemetry: volumes, patterns, peak times



Sit with the agents

Watch the workarounds; the unofficial fixes are the real spec

WHAT DISCOVERY REVEALED IN THE EXAMPLE



"It just stopped working!"

One sentence. Six different realities. Merchants describe symptoms; root causes live in systems.



Connectivity down

Wi-Fi, SIM or router



Card declines

The issuer, not the merchant



Device fault

Hardware or software



Account hold

Risk, KYC or settlement



Platform incident

Everyone's problem, not theirs



Wrong device

POS tablet or printer, not the terminal

DISCIPLINE OF THIS PHASE

Go wide before going deep

Collect every problem type without judging yet. Narrowing too early means solving the loudest problem, not the largest.

Facts over opinions

Every claim gets a source: a call heard, a ticket counted, an expert quoted. Assumptions are written down and marked for validation.

WHERE AI WORKS IN THIS PHASE

AI clusters 10,000 transcripts into themes overnight, quantifies volumes and surfaces outliers no human would find.

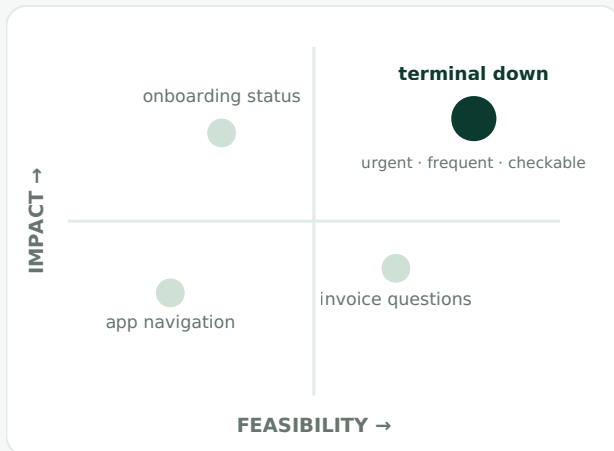
Humans hear the frustration, the workarounds, and what merchants mean but don't say. Analysis is automated; judgment is not.

PHASE 2 · DEFINE

Pick the problem worth solving, and check AI is the right tool

"Solving the wrong problem well is still failure."

PRIORITIZE ON EVIDENCE, NOT ENTHUSIASM



Why "terminal down" wins

Highest urgency (revenue stops mid-queue), high volume, and most root causes are checkable through signals the AI can reach. Impact you can verify, not estimate.

Success defined before building

A resolution is a **processed payment in telemetry within 15 minutes of call end**. Agreed with stakeholders up front, measurable without asking anyone.

THE RIGHT TOOL FOR EACH PART OF THE JOB



Predictable → rules

Ticket routing, status lookups, callbacks. Deterministic automation. No AI needed, no AI used.



Messy language at scale → AI

Understanding a stressed merchant, classifying symptoms, composing signals. This is what takes AI.



Trust and exceptions → humans

Account holds, anger, anything twice-failed. The human path is designed, not a fallback.

THE COMMITMENT THIS PHASE PRODUCES

1

PROBLEM,
FRAMED

One problem statement, one success metric, one scope: **help a verified merchant get from "it stopped working" to taking payments again, and prove it happened**. Everything out of scope is written down and parked, not forgotten.

WHERE AI WORKS IN THIS PHASE

AI sizes the prize: counts affected calls, estimates minutes lost per merchant, simulates conversations to pressure-test the framing. **Humans decide** what gets built and what "good" means. The success metric is a stakeholder decision, never a model output.

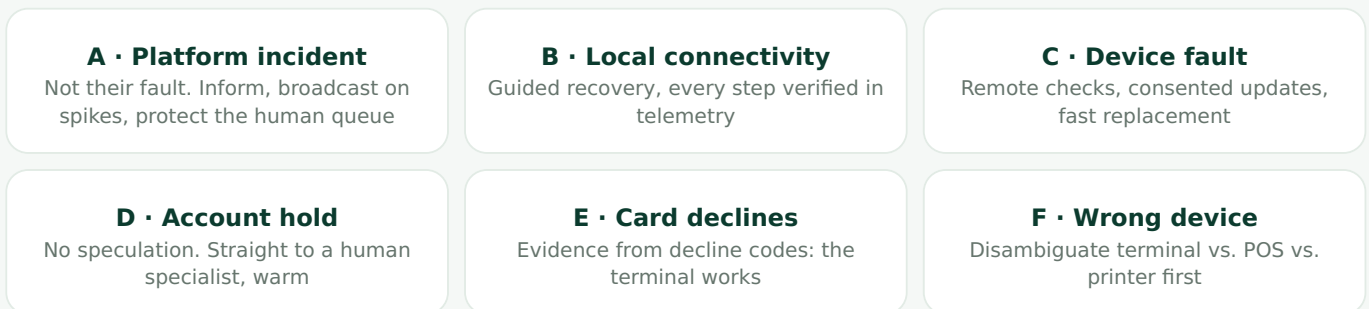
One coherent flow that composes signals, not FAQ answers

"Never ask the merchant what a system already knows."

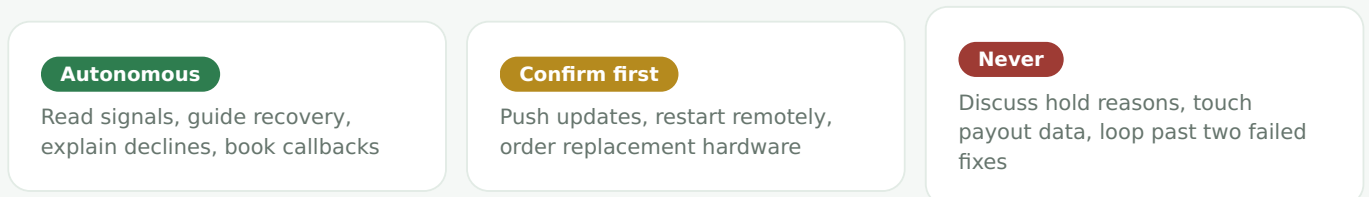
THE FLOW, SIX STEPS



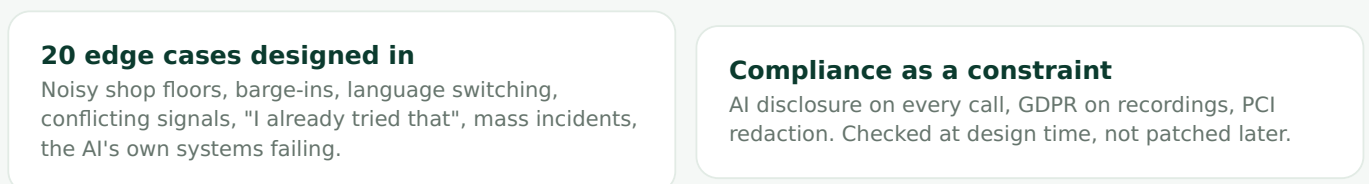
SIX BRANCHES, EACH WITH ITS OWN PLAYBOOK



WHAT THE AI MAY DO



DESIGNED FOR THE WORST CALL, NOT THE AVERAGE ONE



WHERE AI WORKS IN THIS PHASE

The AI is the product here: it listens, classifies, composes signals and speaks. But the control flow stays deterministic: **the model exercises judgment inside steps; rules decide between them.** Reliability comes from the design, not the model.

PHASE 4 · VALIDATE

Prove it works before a merchant ever hears it

"Cheap to kill early, expensive to kill late."

SMALLEST REAL VERSION FIRST



One branch, one market, text first

Connectivity recovery in Denmark as a text simulation, before voice, before five more markets



Learn per iteration, not per launch

Each cycle answers one question: does routing hold? Does recovery verify? Do merchants trust it?

SIX TEST LAYERS BETWEEN DESIGN AND A LIVE CALL

1 · Logic tests

Every branch, gate and edge case as an executable test

2 · Transcript replay

Historic calls through the classifier, scored against known causes

3 · Adversarial callers

Furious, rambling, silent, wrong-device: scripted and survived

4 · Voice reliability

Noise, accents, interruptions, latency: measured, not assumed

5 · Shadow mode

AI proposes on live calls, humans act. Agreement is the gate

6 · Staged pilot

Lowest-risk branches first; account holds never launch autonomous

THE HONESTY GATE

Defined kill criteria, agreed in advance

If replay agreement or shadow-mode agreement stays under the bar after two redesign cycles, the branch does not launch. Killing a weak workflow in testing costs days; killing it in production costs merchant trust.



WHERE AI WORKS IN THIS PHASE

AI generates the hostile test set: synthetic difficult callers, accent and noise variants, thousands of replayed conversations scored automatically. **Humans set the bar** and make the launch call. The model is tested by machines and judged by people.

Autonomy is earned, one branch and one market at a time

"Trust arrives on foot and leaves on horseback."

THREE GEARS OF AUTONOMY



1 · Shadow

AI suggests, humans act. Zero merchant risk, maximum learning



2 · Assisted

AI acts, humans monitor live and can step in mid-call



3 · Autonomous

Per branch, per market, only where the metrics earned it

ONE WORKFLOW, SIX MARKET OVERLAYS

DK

FI

FR

DE

IT

UK

Localize behavior, not logic. Six forked workflows would drift apart in a quarter; one canonical flow with overlays keeps every improvement global.

What the overlay carries

Language and speech models · product suite (terminal vs. full POS bundle) · fiscal and compliance rules · local self-serve links · escalation queues and hours. Market support leads review every script before go-live.

HANDOVER IS PART OF THE BUILD



Living documentation

Workflows, permissions and edge cases anyone can read and challenge



Trained support teams

Agents know what the AI does, and get full context on every handoff



Guardrails that outlive me

Disclosure, GDPR, PCI and permission tiers enforced in the system, not in memory

THE HARD RULE AT EVERY GEAR

The merchant never repeats themselves

Every escalation is a warm transfer with the full story attached: who they are, what they said, what was checked, what was tried, what was promised. A handoff that loses context is a failure, whatever the dashboard says.

WHERE AI WORKS IN THIS PHASE

AI drafts market variants of scripts and self-serve content for expert review, and monitors rollout metrics in real time.
Humans own go-live: each market's support lead signs off before their merchants hear a single AI call.

Launch is the beginning: measure honestly, fix weekly

"Containment is not resolution."

THE METRIC THAT CANNOT LIE

15 min

A resolution is a processed payment

Seen in telemetry within 15 minutes of call end, for hardware and connectivity fixes. An AI that "handled the call" while the terminal is still down has failed twice: it failed the merchant and hid the failure from the team.

FOUR DIALS WATCHED WEEKLY

Verified resolution

Per branch: which parts earn more autonomy

Escalation reasons

Coded, ranked: they are the build list

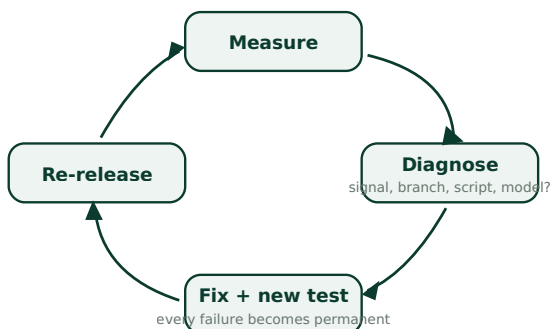
7-day repeats

The detector of fake fixes that bounce back

Trust, AI vs. human

Satisfaction on AI-closed calls must hold the line

THE WEEKLY LOOP



Failures get a name

Missing signal, missing branch, bad script, or model error. Each class has a different fix, and naming it stops the team from blaming "the AI" as a whole.

Invest where the math says

Next branches and new data sources ranked by calls affected x resolution lift. The escalation log, not opinions, writes the roadmap.

WHY THIS METHOD, IN ONE SENTENCE

Discover widely, define sharply, design for the worst call, prove before launch, scale what earns it, and let honest metrics write the roadmap.

Built without inside knowledge of Flatpay's systems: every dependency on internals is a marked assumption with a named validation path, ready to be corrected in week one through the expert interviews this role runs on. The full workflow specification behind page 4 exists and is ready to discuss.

WHERE AI WORKS IN THIS PHASE

AI classifies every failed call and drafts the weekly review; **humans decide** what ships. The same discipline that started this document ends it: AI for scale, judgment for decisions.